

WASP-11b

R-0671

Benchmark run · workflow W8-benchmark-deep · record deep-bench_HATP-10_251211 · created 2026-07-27T18:33:35+00:00

PRACTICE / VALIDATION RUN — not submitted to any science database

Results

Mid-Transit Time (Tmid)	2461020.8953 +/- 0.0038 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.173 +/- 0.051
Transit depth (Rp/Rs)^2	2.99 +/- 1.76 [%]
Orbital Inclination (inc)	89.27 +/- 1.71
Airmass coefficient 1 (a1)	1.0 +/- 0.66
Airmass coefficient 2 (a2)	-0.06 +/- 0.1
Scatter in the residuals of the lightcurve fit is	63.67 %
Best Comparison Star	#3 - [439, 244]
Optimal Aperture	3.0
Optimal Annulus	5.5
Transit Duration (day)	0.105 +/- 0.011

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.173 ± 0.051 vs 0.131 (deviation 0.82σ)
Duration fit vs published	0.105 d vs 0.1075 d (deviation 0.23σ)
Mid-time O–C	-2.9 min
Depth SNR	3.4
Residual scatter	63.67 %

Light curve

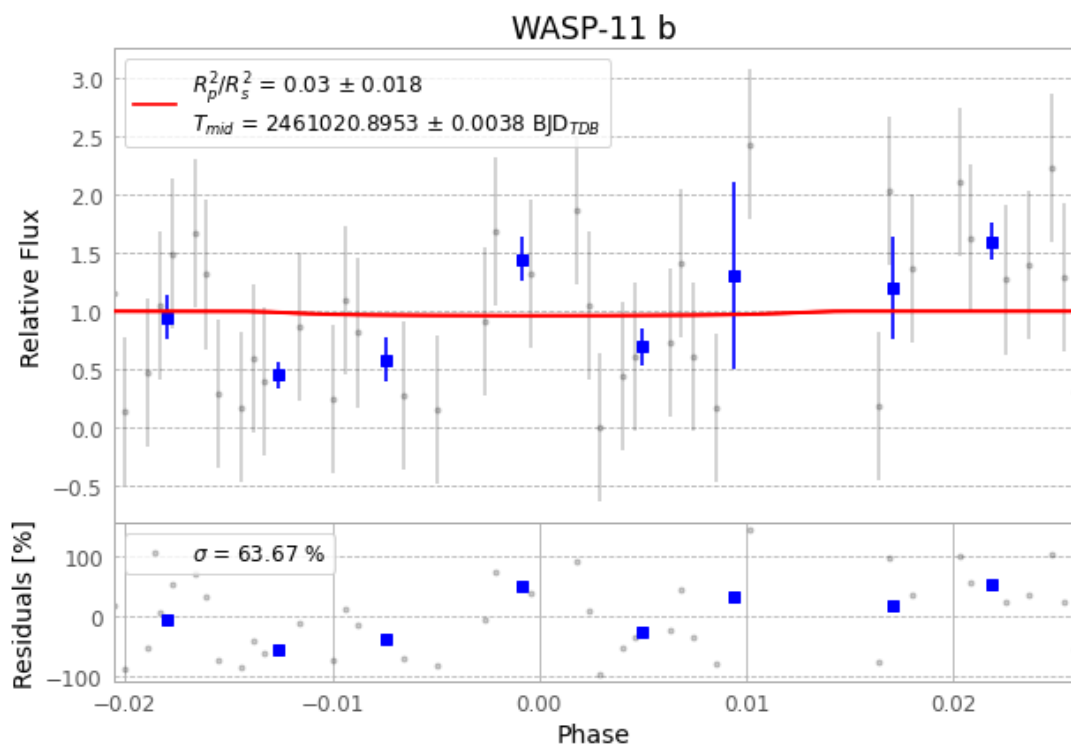


Figure 1 — FinalLightCurve_WASP-11 b_2025-12-11.png (SHA-256 50e61618998712fc...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [400, 239], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 bb23fdb7bdfc209c...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit fc0f9c4

Data provenance

85 FITS frames (56 MB), HATP-10251211073016.FITS → HATP-10251211114215.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-HATP-10-251211.log (d3151bdb8e88...), FinalLightCurve_WASP-11 b_2025-12-11.png (50e616189987...), FinalLightCurve_WASP-11 b_2025-12-11.csv (b0a3f28daed5...)

Compute resources

Wall time 130.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-27 18:33Z.